

Supplementary Information

Semantic-Guided Differentiable Hierarchical Temporal Graph Learning for Class-Imbalanced Log Anomaly Detection

Caizhong Que^{1,2}, Guofang Dong^{1,2,*}

¹ School of Electrical and Information Engineering, Yunnan Minzu University, Kunming 650504, China

² Key Laboratory of Unmanned Autonomous Systems of Yunnan Province, Yunnan Minzu University,
Kunming 650504, China

* Corresponding author: dongguofangyx@163.com

This file contains 13 supplementary tables, two supplementary figures, two short supplementary texts and the reference list for the literature values quoted in Supplementary Table S10. Per-seed metrics for all 724 training runs are provided in the separate workbook *Source Data 1*; the code that produced every item is available in the accompanying repository.

- S1 Statistical testing protocol
- S2 Additional metrics for the main comparison
- S3 Seed-level results for imbalance learning and prototype configurations
- S4 Final hyper-parameters and implementation settings
- S5 Hyper-parameter sensitivity analysis
- S6 Semantic and entity shortcut ablations
- S7 Robustness to parsing noise
- S8 Entity identifiability and entity holdout
- S9 Masked-template pretraining baseline
- S10 Literature reference values and protocol differences
- S11 Baseline taxonomy and comparison protocol
- S12 Negative-sampling strategies for contrastive pretraining
- S13 Semantic field sources and leakage controls
- Supplementary Figures S1–S2
- Supplementary Text S1 Experimental design and reproducibility protocol
- Supplementary Text S2 Data provenance and file hashes

Supplementary Text S1. Experimental design and reproducibility protocol

Data and leakage control. Every dataset is parsed with Drain (depth 4, similarity 0.5) fitted on training-split events only and then frozen; validation and test templates that were not seen in training map to a placeholder. The semantic dictionaries for action and state are fixed before any label is inspected, and no test-set template, anomaly label, large language model or external anomaly knowledge base contributes to them. Where a dataset has an official session key (HDFS block, OpenStack instance, SSH derived key) that key defines the outer sample; otherwise an adaptive idle-gap threshold, estimated from the training partition alone, is used. Splits are 60/20/20; BGL and Thunderbird use a session-stratified random split with a fixed seed, HDFS and OpenStack use a time-ordered split (HDFS groups sessions that share a raw event to prevent replication leakage), and a time-ordered BGL cache is kept for the template-evolution stress test.

Model selection and statistics. Five seeds (42, 123, 256, 512, 1024) are used for the principal experiments. Models are selected on validation AUPRC with early stopping (patience 10) and the decision threshold is calibrated on the validation split by maximising F1; the test split is never used for selection. Comparative claims are supported by paired Wilcoxon signed-rank tests on matched seeds with Holm correction inside each comparison family, reported together with Cliff’s δ ; because five paired seeds give a smallest attainable two-sided p of 0.0625, the paper interprets effect sizes and cross-dataset consistency rather than a binary significance threshold (Supplementary Table S1).

Reproducibility. Each run writes a manifest with the SHA-256 hashes of the raw data, the processed cache and the configuration files, together with the seed and the effective model configuration; checkpoints store the optimiser state and the random-number-generator state. The scripts that produce every table and figure are listed in the repository, and the exact package versions are pinned.

Supplementary Text S2. Data provenance and file hashes

Processed caches used for the controlled experiments:

Dataset	Cache built (UTC)	Code revision	sessions.parquet SHA-256
bgl	2026-08-29T12:14:17	a35db7ff0c	2a9cb2589291bed7...
hdfs	2026-08-29T11:32:16	a35db7ff0c	439dc81b2e177bed...
openstack	2026-08-29T11:05:47	a35db7ff0c	59c7834426ff7176...
ssh	2026-08-29T11:06:54	a35db7ff0c	aa09e7268eadf0a1...
thunderbird	2026-08-29T12:35:04	a35db7ff0c	f246c7d2dfd46d68...

The full hashes, the raw-file hashes and the per-run manifests are included in the repository under `data/processed/*/manifest.json` and `outputs/*/main/**/training_manifest.json`.

Supplementary Table S1. Paired statistical comparisons. Wilcoxon signed-rank tests use the same random seed as the pairing unit; q_{Holm} values are corrected within each family and metric, and Cliff’s δ is the effect size (positive = the first method is better). With five paired seeds the smallest attainable two-sided p is 0.0625.

Dataset	Comparison	Metric	Pairs	Δ (pt)	p	q_{Holm}	Cliff’s δ	Direction
<i>primary</i>								
ssh	GRU-flat (L0)	F1	5	+0.50	1.0000	1.0000	-0.04	—
ssh	GRU-flat (L0)	AUPRC	5	+0.69	0.3125	0.6250	+0.28	+
hdfs	TCN	F1	5	+4.02	0.0625	0.3125	+1.00	+
hdfs	Transformer	AUPRC	5	-0.22	0.0625	0.3125	-1.00	—
bgl	GNN-flat	F1	5	+0.02	0.3125	0.9375	+0.36	+
bgl	GNN-flat	AUPRC	5	-0.00	0.1875	0.5625	-0.44	—
openstack	GRU-flat (L0)	F1	5	-0.00	1.0000	1.0000	-0.12	—
openstack	GRU-flat (L0)	AUPRC	5	-0.00	1.0000	1.0000	-0.04	—
thunderbird	GNN-flat	F1	5	-0.30	0.0625	0.3125	-1.00	—
thunderbird	GNN-flat	AUPRC	5	-0.09	0.1250	0.5000	-0.68	—
<i>vs GRU-flat (L0)</i>								
ssh	GRU-flat (L0)	F1	5	+0.50	1.0000	1.0000	-0.04	—
ssh	GRU-flat (L0)	AUPRC	5	+0.69	0.3125	1.0000	+0.28	+
hdfs	GRU-flat (L0)	F1	5	+5.41	0.0625	0.3125	+1.00	+
hdfs	GRU-flat (L0)	AUPRC	5	-0.10	0.4375	1.0000	-0.52	—
bgl	GRU-flat (L0)	F1	5	+0.06	0.1875	0.7500	+0.76	+
bgl	GRU-flat (L0)	AUPRC	5	+0.00	0.4375	1.0000	+0.44	+
openstack	GRU-flat (L0)	F1	5	-0.00	1.0000	1.0000	-0.12	—
openstack	GRU-flat (L0)	AUPRC	5	-0.00	1.0000	1.0000	-0.04	—
thunderbird	GRU-flat (L0)	F1	5	-0.00	1.0000	1.0000	+0.08	+
thunderbird	GRU-flat (L0)	AUPRC	5	+0.05	0.3125	1.0000	+0.28	+
<i>module step</i>								
hdfs	L7 vs L6	F1	5	+9.89	0.0625	0.3750	+1.00	+
hdfs	L7 vs L0	F1	5	+5.41	0.0625	0.3750	+1.00	+
hdfs	/ L7b vs L7	F1	5	-0.86	0.1875	0.5625	-0.28	—
ssh	L7 vs L6	F1	5	+1.39	0.0625	0.3750	+0.36	+
ssh	L7 vs L0	F1	5	+0.50	1.0000	1.0000	-0.04	—
ssh	/ L7b vs L7	F1	5	-1.47	0.2500	0.5625	-0.40	—

Supplementary Table S2. Additional metrics for the main comparison (mean $\pm$ standard deviation over the five matched random seeds). MCC and balanced accuracy are computed from the reported precision, recall and the test-set class composition.

Dataset	Method	Seeds	AUPRC	AUROC	Precision	Recall	F1	MCC	Balanced accuracy
bgl	TCN	5	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9979 $\pm$ 0.0015	0.9989 $\pm$ 0.0009	0.9984 $\pm$ 0.0009	0.9981 $\pm$ 0.0010	0.9992 $\pm$ 0.0005
bgl	Transformer	5	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9975 $\pm$ 0.0015	0.9994 $\pm$ 0.0004	0.9985 $\pm$ 0.0008	0.9981 $\pm$ 0.0010	0.9994 $\pm$ 0.0003
bgl	GNN-flat	5	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9980 $\pm$ 0.0008	0.9991 $\pm$ 0.0004	0.9986 $\pm$ 0.0003	0.9983 $\pm$ 0.0004	0.9993 $\pm$ 0.0002
bgl	GRU-flat (L0)	5	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9980 $\pm$ 0.0005	0.9983 $\pm$ 0.0007	0.9981 $\pm$ 0.0006	0.9978 $\pm$ 0.0007	0.9989 $\pm$ 0.0004
bgl	SDHTG (L7)	5	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9979 $\pm$ 0.0010	0.9996 $\pm$ 0.0003	0.9987 $\pm$ 0.0004	0.9985 $\pm$ 0.0005	0.9996 $\pm$ 0.0001
hdfs	TCN	5	0.9988 $\pm$ 0.0009	1.0000 $\pm$ 0.0000	0.9113 $\pm$ 0.0231	0.9989 $\pm$ 0.0011	0.9530 $\pm$ 0.0124	0.9533 $\pm$ 0.0121	0.9987 $\pm$ 0.0005
hdfs	Transformer	5	0.9996 $\pm$ 0.0002	1.0000 $\pm$ 0.0000	0.9069 $\pm$ 0.0608	0.9874 $\pm$ 0.0243	0.9439 $\pm$ 0.0228	0.9447 $\pm$ 0.0220	0.9929 $\pm$ 0.0117
hdfs	GNN-flat	5	0.9867 $\pm$ 0.0176	0.9998 $\pm$ 0.0003	0.8295 $\pm$ 0.0642	0.9996 $\pm$ 0.0008	0.9056 $\pm$ 0.0375	0.9087 $\pm$ 0.0352	0.9983 $\pm$ 0.0005
hdfs	GRU-flat (L0)	5	0.9984 $\pm$ 0.0016	1.0000 $\pm$ 0.0000	0.8867 $\pm$ 0.0430	0.9990 $\pm$ 0.0014	0.9391 $\pm$ 0.0234	0.9401 $\pm$ 0.0225	0.9986 $\pm$ 0.0005
hdfs	SDHTG (L7)	5	0.9974 $\pm$ 0.0019	0.9997 $\pm$ 0.0006	0.9955 $\pm$ 0.0046	0.9910 $\pm$ 0.0040	0.9932 $\pm$ 0.0023	0.9931 $\pm$ 0.0023	0.9954 $\pm$ 0.0020
openstack	TCN	5	0.9998 $\pm$ 0.0000	0.9997 $\pm$ 0.0000	0.9991 $\pm$ 0.0020	0.9930 $\pm$ 0.0024	0.9960 $\pm$ 0.0010	0.9913 $\pm$ 0.0022	0.9960 $\pm$ 0.0010
openstack	Transformer	5	0.9999 $\pm$ 0.0001	0.9999 $\pm$ 0.0001	0.9956 $\pm$ 0.0044	0.9974 $\pm$ 0.0024	0.9965 $\pm$ 0.0025	0.9923 $\pm$ 0.0055	0.9960 $\pm$ 0.0029
openstack	GNN-flat	5	0.9999 $\pm$ 0.0001	0.9999 $\pm$ 0.0001	0.9905 $\pm$ 0.0064	0.9982 $\pm$ 0.0039	0.9943 $\pm$ 0.0019	0.9875 $\pm$ 0.0043	0.9933 $\pm$ 0.0025
openstack	GRU-flat (L0)	5	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9983 $\pm$ 0.0039	1.0000 $\pm$ 0.0000	0.9991 $\pm$ 0.0020	0.9981 $\pm$ 0.0043	0.9989 $\pm$ 0.0024
openstack	SDHTG (L7)	5	1.0000 $\pm$ 0.0001	1.0000 $\pm$ 0.0001	0.9991 $\pm$ 0.0020	0.9991 $\pm$ 0.0020	0.9991 $\pm$ 0.0012	0.9981 $\pm$ 0.0026	0.9990 $\pm$ 0.0013
ssh	TCN	5	0.9729 $\pm$ 0.0272	0.9971 $\pm$ 0.0019	0.8781 $\pm$ 0.0251	0.9944 $\pm$ 0.0124	0.9324 $\pm$ 0.0130	0.9234 $\pm$ 0.0143	0.9861 $\pm$ 0.0055
ssh	Transformer	5	0.7484 $\pm$ 0.0905	0.9283 $\pm$ 0.0291	0.7966 $\pm$ 0.0715	0.7000 $\pm$ 0.2689	0.7132 $\pm$ 0.2068	0.6960 $\pm$ 0.1788	0.8348 $\pm$ 0.1280
ssh	GNN-flat	5	0.4171 $\pm$ 0.0609	0.8356 $\pm$ 0.0229	0.5052 $\pm$ 0.0445	0.6444 $\pm$ 0.0795	0.5651 $\pm$ 0.0500	0.4913 $\pm$ 0.0598	0.7713 $\pm$ 0.0382
ssh	GRU-flat (L0)	5	0.9854 $\pm$ 0.0122	0.9920 $\pm$ 0.0111	0.9766 $\pm$ 0.0131	0.9278 $\pm$ 0.0248	0.9515 $\pm$ 0.0168	0.9444 $\pm$ 0.0190	0.9621 $\pm$ 0.0128
ssh	SDHTG (L7)	5	0.9922 $\pm$ 0.0031	0.9985 $\pm$ 0.0006	0.9943 $\pm$ 0.0128	0.9222 $\pm$ 0.0362	0.9565 $\pm$ 0.0183	0.9510 $\pm$ 0.0201	0.9607 $\pm$ 0.0178
thunderbird	TCN	5	0.9996 $\pm$ 0.0001	1.0000 $\pm$ 0.0000	0.9997 $\pm$ 0.0004	0.9943 $\pm$ 0.0017	0.9970 $\pm$ 0.0007	0.9970 $\pm$ 0.0007	0.9971 $\pm$ 0.0008
thunderbird	Transformer	5	0.9994 $\pm$ 0.0006	0.9999 $\pm$ 0.0003	0.9988 $\pm$ 0.0026	0.9960 $\pm$ 0.0022	0.9974 $\pm$ 0.0023	0.9974 $\pm$ 0.0023	0.9980 $\pm$ 0.0011
thunderbird	GNN-flat	5	0.9998 $\pm$ 0.0001	1.0000 $\pm$ 0.0000	0.9978 $\pm$ 0.0006	0.9978 $\pm$ 0.0006	0.9989 $\pm$ 0.0003	0.9989 $\pm$ 0.0003	0.9989 $\pm$ 0.0003
thunderbird	GRU-flat (L0)	5	0.9983 $\pm$ 0.0001	0.9996 $\pm$ 0.0004	0.9996 $\pm$ 0.0006	0.9922 $\pm$ 0.0009	0.9959 $\pm$ 0.0005	0.9959 $\pm$ 0.0005	0.9961 $\pm$ 0.0005
thunderbird	SDHTG (L7)	5	0.9989 $\pm$ 0.0008	0.9996 $\pm$ 0.0005	0.9997 $\pm$ 0.0004	0.9921 $\pm$ 0.0012	0.9959 $\pm$ 0.0006	0.9959 $\pm$ 0.0006	0.9960 $\pm$ 0.0006

Supplementary Table S3. Imbalance-learning and prototype configurations on SSH. Each configuration is trained with the same protocol as the main experiments (30 epochs, three seeds); the corresponding cross-dataset incremental ladder is reported in the main text.

Configuration	Seeds	AUPRC	F1
bce	3	0.9720 ± 0.0156	0.9328 ± 0.0371
focal	3	0.9874 ± 0.0154	0.9612 ± 0.0428
multi_prototype	3	0.9939 ± 0.0024	0.9508 ± 0.0314
multi_prototype_div	3	0.9932 ± 0.0025	0.9664 ± 0.0229
no_prototype	3	0.9882 ± 0.0055	0.9372 ± 0.0084
single_prototype	3	0.9931 ± 0.0031	0.9508 ± 0.0314
weighted_bce	3	0.9845 ± 0.0147	0.9521 ± 0.0349

Supplementary Table S4. Final hyper-parameters and implementation settings, with the configuration key that carries each value. Parameters without a sensitivity scan are fixed at the stated value.

Symbol	Meaning	Final value	Sensitivity scan	Verified against
K	number of normal prototypes	8	1, 2, 4, 16	configs/model/sdhtg.yaml: num_normal_prototypes
tau_p	prototype soft-assignment temperature	0.10	0.05, 0.20, 0.50	configs/model/sdhtg.yaml: prototype_temperature; sens_prototype_temperature_*
lambda_p	prototype-distance scale	learned (softplus)	fixed at 0, 0.25, 0.5, 1.0, 2.0	prototype_scale_override defaults to null; sens_prototype_scale_*
m	prototype margin	0.5	not scanned (code default family)	configs/experiment/*.yaml: loss.prototype_margin
rho_p	prototype similarity threshold	0.20 (code default)	not scanned	src/sdhtg/models/config.py default
tau_b	boundary temperature (annealed 1.0 -> value)	0.10	final value 0.05, 0.25, 0.50, 1.00	configs/model/sdhtg.yaml: boundary.final_temperature
(r_A, r_E)	target action / entity boundary rates	0.05 / 0.01	0.01/0.005, 0.01/0.001, 0.10/0.02, 0.10/0.05	configs/experiment/*.yaml: loss.*_boundary_rate
R_l	local temporal radius (status/action/entity)	8 / 4 / 2	halved (4/2/1), doubled (16/8/4)	configs/model/sdhtg.yaml: local_temporal_radius
K_l	same-semantic neighbours (status/action/entity)	4 / 4 / 2	halved (2/2/1), doubled (8/8/4)	configs/model/sdhtg.yaml: semantic_neighbors
eps_m	membership-edge threshold	1e-6	1e-8, 1e-4, 1e-2	configs/model/sdhtg.yaml: hierarchy.membership_epsilon
gamma	focal modulation exponent	2.0	not scanned	configs/experiment/*.yaml: loss.focal_gamma
beta	effective-number weighting exponent	0.9999	not scanned	configs/experiment/*.yaml: loss.effective_number_beta
w_proto / w_div / w_bal	prototype, diversity, balance loss weights	0.05 / 0.1 / 0.1	diversity+balance removed in the final model	configs/experiment/*.yaml: loss.prototype_*_weight
–	masked-template pre-training (LogBERT route, supplementary baseline)	15% mask, 10 epochs	not scanned	scripts/train_mlm_baseline.py

Supplementary Table S5. Hyper-parameter sensitivity scan on SSH. Each variant is trained under the default configuration with one parameter changed; the reference is the default configuration evaluated on the same seeds (validation AUPRC 0.9853, test AUPRC 0.9941, test F1 0.9561). Per-seed values are provided in Source Data 1.

Setting (group_variant)	Seeds	Validation AUPRC	Test AUPRC	Test F1	Δ F1 vs default (pt)
boundary_rate_ra001_re0001	3	0.9857	0.9935	0.9664	+1.03
boundary_rate_ra001_re0005	3	0.9816	0.9960	0.9710	+1.49
boundary_rate_ra010_re0020	3	0.9850	0.9941	0.9515	-0.46
boundary_rate_ra010_re0050	3	0.9853	0.9934	0.9515	-0.46
boundary_temperature_tau005	3	0.9858	0.9938	0.9521	-0.40
boundary_temperature_tau025	3	0.9858	0.9930	0.9619	+0.58
boundary_temperature_tau050	3	0.9865	0.9941	0.9561	-0.00
boundary_temperature_tau100	3	0.9853	0.9934	0.9470	-0.91
membership_epsilon_eps1e-4	3	0.9856	0.9931	0.9515	-0.46
membership_epsilon_eps1e-8	3	0.9829	0.9947	0.9664	+1.03
prototype_scale_lambda000	3	0.9811	0.9858	0.9521	-0.40
prototype_scale_lambda025	3	0.9838	0.9915	0.9475	-0.86
prototype_scale_lambda050	3	0.9855	0.9937	0.9572	+0.11
prototype_scale_lambda100	3	0.9846	0.9943	0.9622	+0.61
prototype_scale_lambda200	3	0.9802	0.9960	0.9814	+2.53
prototype_temperature_tau005	3	0.9845	0.9945	0.9616	+0.55
prototype_temperature_tau020	3	0.9864	0.9947	0.9515	-0.46
prototype_temperature_tau050	3	0.9855	0.9943	0.9612	+0.51
prototypes_k1	3	0.9852	0.9926	0.9410	-1.51
prototypes_k16	3	0.9856	0.9940	0.9561	-0.00
prototypes_k2	3	0.9855	0.9932	0.9365	-1.96
prototypes_k4	3	0.9856	0.9930	0.9514	-0.47
semantic_neighbors_double	3	0.9824	0.9951	0.9665	+1.04
semantic_neighbors_half	3	0.9830	0.9939	0.9566	+0.05
temporal_radius_double	3	0.9849	0.9941	0.9508	-0.53
temporal_radius_half	3	0.9857	0.9935	0.9417	-1.44

Supplementary Table S6. Semantic and entity shortcut controls on SSH: each row removes or alters one selected cue (entity embedding, boundary priors, action/state sources, entity identity) while keeping the remaining inputs unchanged. The last row repeats the full-input reference.

Variant	Seeds	AUPRC	F1
entity_unk	5	0.9907 $\pm$ 0.0042	0.9479 $\pm$ 0.0249
mask_status_words	5	0.9937 $\pm$ 0.0034	0.9598 $\pm$ 0.0216
no_action_prior	5	0.9929 $\pm$ 0.0029	0.9538 $\pm$ 0.0193
no_action_source	5	0.9917 $\pm$ 0.0033	0.9449 $\pm$ 0.0265
no_entity_embed	5	0.9930 $\pm$ 0.0030	0.9541 $\pm$ 0.0162
no_entity_prior	5	0.9928 $\pm$ 0.0035	0.9573 $\pm$ 0.0110
no_status_source	5	0.9924 $\pm$ 0.0030	0.9482 $\pm$ 0.0269
shuffle_entity	5	0.9919 $\pm$ 0.0046	0.9600 $\pm$ 0.0192
template_time_only	5	0.9922 $\pm$ 0.0048	0.9568 $\pm$ 0.0313
full input (reference)	5	0.9922 $\pm$ 0.0031	0.9565 $\pm$ 0.0183

Supplementary Table S7. Paired evaluation under template-level parsing noise (SSH). Replacement, merge, split and UNK-injection corruptions are applied at rewrite rates of 20% and 40%; “test_only” corrupts the test split, “all” corrupts training, validation and test. Model parameters and the decision threshold calibrated on the clean validation split are kept fixed, and the clean column re-evaluates the same weights on the clean test split. Δ is in percentage points.

Rate	Corruption	Protocol	Seeds	AUPRC _{noisy}	AUPRC _{clean}	Δ AUPRC	F1 _{noisy}	F1 _{clean}	Δ F1
20%	merge	all	2	0.9926	0.9922	+0.05	0.9412	0.9644	-2.32
20%	merge_test	only	2	0.9929	0.9918	+0.11	0.9420	0.9722	-3.02
20%	replace	all	2	0.9946	0.9940	+0.05	0.9567	0.9650	-0.83
20%	replace_test	only	2	0.9933	0.9902	+0.31	0.9333	0.9334	-0.01
20%	split	all	2	0.9962	0.9966	-0.04	0.9787	0.8959	+8.27
20%	split_test	only	2	0.9958	0.9961	-0.03	0.9563	0.9718	-1.55
20%	unk	all	2	0.9946	0.9943	+0.03	0.9567	0.9650	-0.83
20%	unk_test	only	2	0.9919	0.9920	-0.00	0.9575	0.9291	+2.84
40%	merge	all	2	0.9941	0.9481	+4.60	0.9488	0.4861	+46.27
40%	merge_test	only	2	0.9919	0.9905	+0.14	0.9344	0.9499	-1.55
40%	replace	all	2	0.9944	0.9944	+0.00	0.9420	0.9567	-1.47
40%	replace_test	only	2	0.9949	0.9946	+0.03	0.9567	0.9656	-0.88
40%	split	all	2	0.9944	0.9940	+0.03	0.9571	0.9646	-0.74
40%	split_test	only	2	0.9943	0.9953	-0.10	0.9426	0.9567	-1.41
40%	unk	all	2	0.9925	0.9912	+0.13	0.9429	0.9650	-2.21
40%	unk_test	only	2	0.9932	0.9912	+0.21	0.9644	0.9179	+4.65

Supplementary Table S8. Entity-identifiability evaluation. The entity identities covering about 20% of the test sessions are masked to a placeholder in training, validation and test, and the model is retrained; “seen” and “holdout” then refer to the same weights and the same calibrated threshold. OpenStack has no negatives in the holdout group, so AUPRC is undefined there.

Dataset	Group	Seeds	Samples	Positives	AUPRC	F1	Precision	Recall
bgl	all	2	31688	5280	1.0000	0.9977	0.9969	0.9986
bgl	holdout	2	11530	1580	1.0000	0.9954	0.9934	0.9975
bgl	seen	2	20000	3697	1.0000	0.9987	0.9984	0.9991
openstack	all	2	417	228	0.9999	0.8535	0.7734	1.0000
openstack	holdout	2	169	169	–	1.0000	1.0000	1.0000
openstack	seen	2	248	59	0.9992	0.6922	0.6190	1.0000

Supplementary Table S9. Masked-template pretraining baseline under the controlled input protocol. The baseline shares the inputs, encoder depth and width, supervised objective, early-stopping rule and threshold calibration with the flat Transformer; only the 15% masked-template pretraining stage is added. SSH and OpenStack use five seeds, BGL one (it is saturated).

Dataset	Method	Seeds	AUPRC	F1
bgl	SDHTG	5	1.0000 $\pm$ 0.0000	0.9987 $\pm$ 0.0004
bgl	Transformer (no pretraining)	5	1.0000 $\pm$ 0.0000	0.9985 $\pm$ 0.0008
bgl	Transformer + masked-template pretraining	1	0.9999	0.9985
openstack	SDHTG	5	1.0000 $\pm$ 0.0001	0.9991 $\pm$ 0.0012
openstack	Transformer (no pretraining)	5	0.9999 $\pm$ 0.0001	0.9965 $\pm$ 0.0025
openstack	Transformer + masked-template pretraining	5	0.9994 $\pm$ 0.0013	0.9978 $\pm$ 0.0022
ssh	SDHTG	5	0.9922 $\pm$ 0.0031	0.9565 $\pm$ 0.0183
ssh	Transformer (no pretraining)	5	0.7484 $\pm$ 0.0905	0.7132 $\pm$ 0.2068
ssh	Transformer + masked-template pretraining	5	0.8953 $\pm$ 0.0245	0.7634 $\pm$ 0.0283

Supplementary Table S10. Reference values reported by prior studies under their own protocols. Differences in preprocessing, split construction, label definition, threshold selection and evaluation protocol prevent direct numerical comparison with the controlled results of this work; the table is provided for context only.

Source	Method	HDFS	BGL	Thunderbird	OpenStack	SSH	Metric	Protocol note
original paper	HitAnomaly	0.983	0.921	–	0.862	–	F1	fixed time-window sessions; original Figs. 10/16/11
original paper	LogAnomaly	0.950	0.960	–	–	–	F1	sliding window of 20 blocks/sessions
original paper	LogBERT	0.823	0.908	0.966	–	–	F1	native session sequences; original Table 2
original paper	LAnoBERT	0.965	0.875	0.999	–	–	F1	parser-free masked LM; predictive-probability score; original Table 3
original paper	DeepCASE	0.904	–	–	–	–	F1	event sequences, 80% training data; original Appendix E
original paper	BASN	0.908	0.912	–	0.889	0.871	F1	differentiable soft-reset single-level boundaries, zero-shot; original Table 6
original paper	HLogformer	–	–	–	–	–	F1	evaluated on Cloud-Trail/OKTA/Amazon Reviews, not on LogHub
original paper	Log2vec	–	–	–	–	–	AUC	CERT/LANL insider-threat detection, not LogHub
original paper	DeepTraLog	–	–	–	–	–	F1	TrainTicket microservice traces (F1 0.954), different data type
third-party reproduction	DeepLog	0.773	0.861	0.931	–	–	F1	same-protocol reproduction reported by the LogBERT paper
third-party reproduction	LogAnomaly	0.562	0.741	0.927	–	–	F1	same source; 0.29 lower than its original protocol
this work (controlled protocol)	SDHTG	0.993	0.999	0.996	0.999	0.957	F1	outer samples, five seeds

Supplementary Table S11. Baseline taxonomy and comparison protocol. “Controlled” baselines are implemented and trained in this work under identical inputs, splits, seeds and threshold selection; literature references are quoted and never merged into the controlled comparisons.

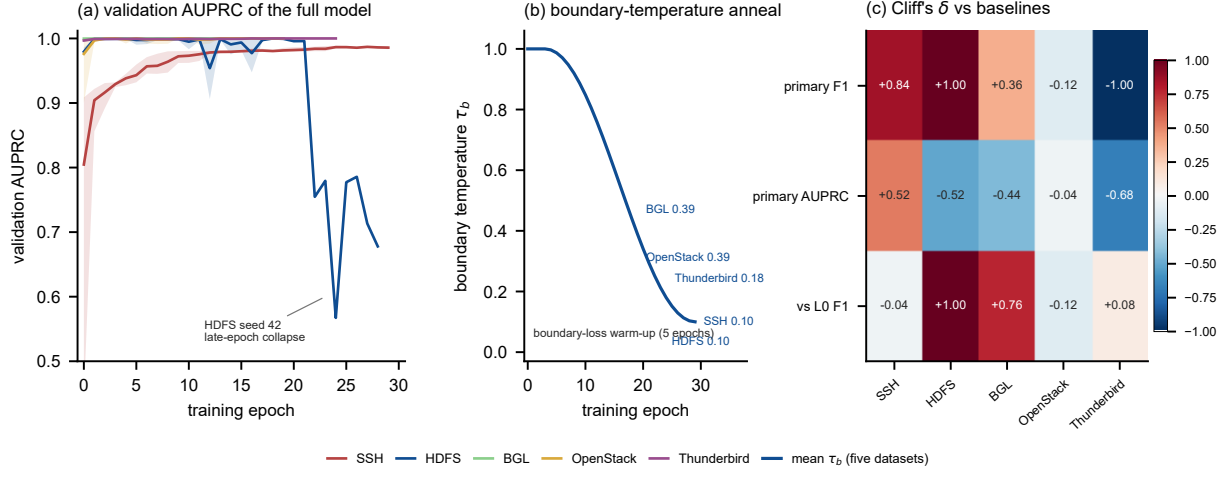
Category	Representative methods	Original protocol	Role in this work
classical sequence	DeepLog, LogAnomaly	session template sequences, next-event and count anomalies	literature reference
pretrained language model	LogBERT, LAnoBERT	masked-template prediction, normal-pattern hypersphere	literature reference
hierarchical	HitAnomaly, HLogformer	hierarchical Transformer over session sequences or nested log entries	literature reference
context construction	DeepCASE	semi-supervised event clustering, two-stage detection	literature reference
graph-based	Log2vec, DeepTraLog	representation learning on log graphs or call graphs	literature reference
boundary-aware	BASN	differentiable soft-reset boundaries, no boundary labels	literature reference (closest related work)
controlled baselines	GRU-flat, TCN, Transformer, GNN-flat	identical template/entity/action/state/time inputs; no learned boundary, no heterogeneous graph	implemented here (main comparison)
imbalance learning	BCE, weighted BCE, focal, CB-focal	same inputs and splits, classification loss / prototype setting replaced	implemented here
masked-template pretraining	Transformer + masked-template prediction	identical inputs and encoder, 15% template masking before supervised tuning	implemented here (supplementary baseline)

Supplementary Table S12. Negative-sampling strategies for contrastive pretraining on SSH. “normal_only” pretrains on normal sessions, “all_train” also uses unlabelled anomalous sessions; the suffix names the negative-selection rule (random, hardest, semi-hard, semantically filtered, no negatives, supervised contrastive). The reference value without pretraining is AUPRC 0.9922 ± 0.0031 .

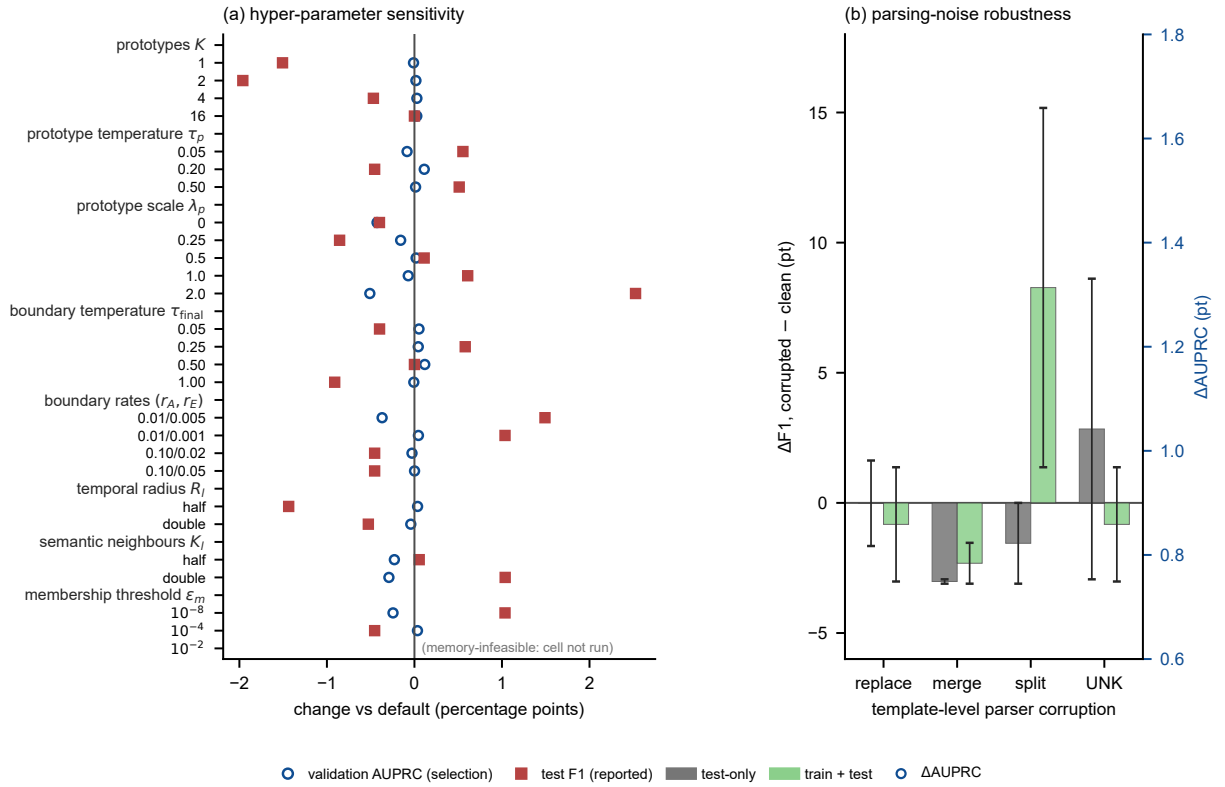
Protocol (sampling __ data range)	Seeds	AUPRC		F1
all_train_hard	3	0.9800 ± 0.0166	0.9333 ± 0.0083	
all_train_none	3	0.9937 ± 0.0023	0.9460 ± 0.0234	
all_train_random	3	0.9830 ± 0.0077	0.9331 ± 0.0100	
all_train_semantic	3	0.9939 ± 0.0026	0.9561 ± 0.0258	
all_train_semi_hard	3	0.9907 ± 0.0051	0.9493 ± 0.0083	
all_train_supervised	3	0.9912 ± 0.0034	0.9365 ± 0.0174	
normal_only_hard	3	0.9789 ± 0.0134	0.9427 ± 0.0151	
normal_only_none	3	0.9918 ± 0.0055	0.9564 ± 0.0151	
normal_only_random	3	0.9765 ± 0.0117	0.9378 ± 0.0088	
normal_only_semantic	3	0.9950 ± 0.0024	0.9612 ± 0.0428	
normal_only_semi_hard	3	0.9887 ± 0.0017	0.9217 ± 0.0064	
normal_only_supervised	3	0.9946 ± 0.0018	0.9566 ± 0.0254	

Supplementary Table S13. Sources of the semantic fields and the controls used to prevent label or test-set leakage. No field is derived from anomaly labels, test-split templates, large language models or external anomaly knowledge bases; unknown values map to placeholders.

Field	Construction	Source of the labels	Leakage control
template	Drain parser (depth 4, similarity 0.5)	fitted on training-split events only, then frozen	test/validation templates unseen in training map to UNK_TEMPLATE
action	deterministic prior dictionary over training template texts	fixed before seeing any labels	no test-set templates, no LLM, no external anomaly knowledge
state	deterministic status dictionary plus template-identity fallback	fixed before seeing any labels	explicit anomaly words (failed/error/timeout) are ablated in Section 6.4
entity	dataset-specific key (node, block id, host, service)	taken from the raw log fields	train-side vocabulary; unseen entities map to a placeholder
time	per-event inter-arrival gap, continuous-time encoding	derived from the timestamps within an outer sample	no cross-split statistics are used
adaptive idle threshold	median inter-event gap per entity x 2 (capped 3600 s)	estimated on the training partition only	validation/test partitions never re-estimate the threshold



Supplementary Figure S1. Training dynamics and paired effect sizes. (a) Validation AUPRC against the training epoch for the five datasets (five seeds; line = mean, band = range). (b) The boundary-temperature anneal recorded during training. (c) Cliff's δ of the paired comparisons against the strongest controlled baseline and against the flat GRU baseline. The HDFS seed that collapses after epoch 23 is visible in (a) and is why model selection relies on validation-based early stopping.



Supplementary Figure S2. Hyper-parameter sensitivity and parsing-noise robustness. (a) Change of validation AUPRC (open circles, selection criterion) and test F1 (filled squares) relative to the default configuration, in percentage points. (b) Paired change under template-level parsing noise at a 20% rewrite rate: bars are $\Delta F1$ (mean, range over two seeds) and open circles are Δ AUPRC. Ranking quality is unchanged while F1 moves in both directions, i.e. the F1 movement reflects threshold calibration rather than a loss of detection ability.

Supplementary References

1. Guo H, Yuan S, Wu X. LogBERT: log anomaly detection via BERT. In: 2021 International Joint Conference on Neural Networks (IJCNN), 2021.
2. Du M, Li F, Zheng G, Srikumar V. DeepLog: anomaly detection and diagnosis from system logs through deep learning. In: Proceedings of the ACM SIGSAC Conference on Computer and Communications Security, 2017: 1285–1298.
3. Meng W, Liu Y, Zhu Y, et al. LogAnomaly: unsupervised detection of sequential and quantitative anomalies in unstructured logs. In: Proceedings of the 28th International Joint Conference on Artificial Intelligence, 2019: 4739–4745.
4. Huang S, Liu Y, Fung C, et al. HitAnomaly: hierarchical transformers for anomaly detection in system log. *IEEE Transactions on Network and Service Management*, 2020, 17(4): 2064–2076.
5. Lee Y, Kim J, Kang P. LAnoBERT: system log anomaly detection based on BERT masked language model. *Applied Soft Computing*, 2023, 146: 110689.
6. Zhang C, Peng X, Sha C, et al. DeepTraLog: trace-log combined microservice anomaly detection through graph-based deep learning. In: Proceedings of the 44th International Conference on Software Engineering (ICSE), 2022: 623–634.
7. Liu F, Wen Y, Zhang D, et al. Log2vec: a heterogeneous graph embedding based approach for detecting cyber threats within enterprise. In: Proceedings of the ACM SIGSAC Conference on Computer and Communications Security, 2019: 1777–1794.
8. Ailabouni F, Román-Gallego J Á, Pérez-Delgado M L, Grande Pérez L. Boundary-aware contrastive learning for log anomaly detection. *Applied Sciences*, 2026, 16(7): 3208.
9. van Ede T, Aghakhani H, Spahn N, et al. DeepCASE: semi-supervised contextual analysis of security events. In: Proceedings of the IEEE Symposium on Security and Privacy, 2022: 522–539.
10. Hou Z, Ghashami M, Kuznetsov M, Torkamani M. HLogformer: a hierarchical transformer for representing log data. *arXiv preprint arXiv:2408.16803*, 2024.
11. He P, Zhu J, Zheng Z, Lyu M R. Drain: an online log parsing approach with fixed depth tree. In: Proceedings of the IEEE International Conference on Web Services (ICWS), 2017: 33–40.